AIMScribe v3 - Failure Review

Every way the system can fail, what it does today, the evidence, and what to fix next. Reviewed against the code as of 16 October 2026.

Verdict	Count	Meaning
HANDLED	32	Handled, with a test or a live run as evidence
PARTIAL	15	Handled in part, or depends on CMED or the clinic
GAP	4	Not handled: audio can be lost, misfiled or stuck
Total	51	

The ten that matter most

#	Failure	Consequence	Fix
1. U1	Laptop clock behind the server by more than 5 s	Nothing from that PC reaches the server; audio piles up on the laptop	Verify iat/exp against the server's Date header offset (or leeway 10 min), report clock skew in the heartbeat, and have the installer enable Windows time sync *(half a day)*
2. D1	A migration not applied on a deployed database	Uploads stall until someone notices; recorders retry for ever	A schema_version table written by each migration and checked at start (refuse to start or report degraded in /health) *(half a day)*
3. A3	The droplet is lost	Up to a week of clinical records and the catalogue lost (DigitalOcean weekly backup only)	Upload each encrypted nightly backup to its own R2 bucket *(half a day)*
4. B3	CMED's finished page meets the new recorder	If API 2 is missing for a click, that part is archived unconfirmed (kept)	Run tools/cmed_build_session.py with CMED's build *(1 session)*
5. S9	Doctor's personal time after a no-prescription visit	Up to 15 min of the doctor's own time in a follow-up's file	Pause button for personal calls; trim trailing silence on the server *(half a day)*
6. S4	Recorder crashes mid-consultation	Second part archived as its own file, not joined to the first	Carry 'continues: <session>' in the new session and let the server join them *(1 day)*
7. S10	Cases the rules cannot settle	A person must look at the flagged file	Review list on the dashboard *(half a day)*
8. C2	Laptop disk full (C: drive) while recording	Audio after the failure is lost while the recording looks alive	Catch write errors in the segmenter loop, keep the PCM in memory and retry, raise *disk_full*; check free disk space as well as spool size *(1 day)*
9. D4	A CMED outage makes many recordings unconfirmed	Alert flood during every CMED outage hides real alerts	Raise one alert per clinic per outage, or only after 24 h without API 2 *(2 hours)*
10. U3	Server says no (unmapped clinic, revoked device, key mismatch)	Revoked device holds audio until the spool fills; then new starts are refused	Dashboard list of held sessions per PC with the reason *(half a day)*

Most urgent: U1. A laptop clock only six seconds behind the server makes every grant fail, so nothing from that PC is ever uploaded. Since today's change those recordings are held rather than deleted, so nothing is lost - but nothing arrives either. Clinic laptops drift; this will happen.

PART 1

Every failure mode, by component

What happens today, with the code it was checked against and the evidence.

Recorder - capture

HANDLED	C1 - Microphone unplugged or headset switched mid-consultation
Today	A watchdog notices within about 2.5 s - failed reads, or the read that never returns when a USB microphone is pulled out - and reopens the Windows default microphone in the same session (core/recorder.py _watch, _failover). The doctor sees 'Microphone changed. Still recording'; the server gets *microphone_switched* with the seconds lost.
Evidence	tests/test_microphone_failover.py (8, fake PortAudio incl. a read that hangs); not tried with a real unplug
Consequence	About 2-3 s of audio at the moment of the switch
Fix	-

GAP	C2 - Laptop disk full (C: drive) while recording
Today	A failed segment write still ends the segmenter thread (core/simple_splitter.py). Unlikely in practice: audio is deleted from the PC once the server holds it, and a start is refused when the spool has under 4 minutes of room.
Evidence	Code read
Consequence	Audio after the failure is lost while the recording looks alive
Fix	Catch write errors in the segmenter loop, keep the PCM in memory and retry, raise *disk_full*; check free disk space as well as spool size *(effort: 1 day)*

HANDLED	C3 - Laptop sleeps or the lid is closed mid-consultation
Today	While a recording is open the recorder tells Windows the PC is in use, so idle sleep cannot happen (core/power.py). The installer sets 'lid close: do nothing' and turns off USB selective suspend, which could power the speakerphone down.
Evidence	tests/test_power.py; installer compiled; not tried on a real laptop
Consequence	-
Fix	-

HANDLED	C4 - Power cut or forced shutdown mid-recording
Today	The unsealed part of the current clip is written to disk, encrypted, every 3 s (spool partial.aimspl). On the next start it is sealed into the chain as the last segment. Segments themselves are fsynced and renamed; a torn journal line is repaired.
Evidence	tests/test_partial_audio.py (7), tests/test_spool.py
Consequence	At most about 3 s of audio
Fix	-

HANDLED	C5 - Room is silent / microphone muted
Today	Two silent segments raise *silent_session* once; it reaches the dashboard live or from the health log. The clinic microphone is kept at 100% and unmuted (see C7).
Evidence	Live outage test: silent_session arrived from the health log
Consequence	Alert only; a muted session still records nothing
Fix	-

HANDLED	C6 - Windows shutdown, restart or sign-out mid-consultation
Today	A hidden window receives the shutdown notice, holds it for up to 4 s with 'AIMScribe is saving the consultation', closes the recording cleanly (reason *pc_shutdown*) and lets Windows continue (core/shutdown_guard.py). Shutdown is never refused.
Evidence	tests/test_shutdown_guard.py (9, incl. the real hidden window); no real shutdown run
Consequence	-
Fix	-

HANDLED	C7 - Jabra speakerphone not the Windows default, or turned down
Today	The recorder opens the microphone whose name contains 'Jabra' whatever the default is, and keeps its level at 100% and unmuted at start-up and on every idle heartbeat (core/microphone.py). Without it the PC's own microphone records.
Evidence	tests/test_microphone_failover.py (Jabra tests); level read and written live on this PC's microphone; no real Jabra tried
Consequence	-
Fix	-

Recorder - start and labelling

HANDLED	S1 - Wrong patient opened first, then the right one
Today	Every click while the recording is not armed is accepted (RECORDING_STARTED, same recording) and signed into the chain with its audio second. The server applies R3: a first patient open under 60 s, never returned to and never prescribed, belongs to the next. One file under the right patient.
Evidence	Live: Rahim then Karim at 8 s -> one file TEST_KARIM (R3+R1); backend tests/test_labelling.py
Consequence	-
Fix	-

HANDLED	S2 - Recorder crashes after a click, before the close
Today	Click and prescription markers are journalled and delivered in chain order after the restart; the server labels from them as usual.
Evidence	tests/test_start_recovery.py (crash after a click), tests/test_markers.py
Consequence	-
Fix	-

HANDLED	**S3 - Microphone busy at the moment CMED triggers**
Today	Start is kept and retried every 5 s for 15 min until it records, or until CMED sends another patient.
Evidence	tests/test_start_recovery.py
Consequence	Seconds of audio at the start
Fix	-

PARTIAL	**S4 - Recorder crashes mid-consultation**
Today	Interrupted session closed on restart; a consultation live within 10 min is recorded again under the same five fields. Two sessions then share one visit and are not tied to each other.
Evidence	tests/test_start_recovery.py
Consequence	Second part archived as its own file, not joined to the first
Fix	Carry 'continues: <session>' in the new session and let the server join them *(effort: 1 day)*

PARTIAL	**S5 - Recorder not running when the doctor opens the patient**
Today	The trigger never arrives. The startup task now restarts a stopped recorder within a minute; CMED's page has no resend, so that consultation is not recorded.
Evidence	Installer task: logon trigger repeating every minute + restart on failure
Consequence	Consultation not recorded if the recorder was down at the click
Fix	Nothing further on our side without CMED; keep the recorder running (it is)

HANDLED	**S6 - Follow-up with no prescription, then the next patient**
Today	The click on the next patient is accepted and logged; the server applies R4: a patient open 60 s or more, not prescribed and not returned from, is a real consultation. The recording is cut at that click into two files.
Evidence	Live: Sakib 70 s, Salma clicked, her prescription -> two files, Sakib 0-66.4 s (R4) and Salma (R1); the two rejoin to the signed hash
Consequence	-
Fix	-

HANDLED	**S7 - Recording never stopped**
Today	A recording ends at the next click after a prescription; 2 min (+ up to 10 s, at a pause) after the prescription; after 15 min with no CMED activity; at Windows shutdown; or with Stop.
Evidence	Live: closed by itself 121 s after the prescription (counselling_limit); tests/test_markers.py (both limits)
Consequence	-
Fix	-

HANDLED	**S8 - Same trigger sent twice / page reload**
Today	Before the prescription: accepted, same recording, no extra marker. After it: SESSION_ALREADY_ACTIVE. A repeated prescription_built: GATE_ALREADY_ARMED.
Evidence	tests/test_channel_a.py, tools/channel_a_test.py 24/24
Consequence	-
Fix	-

PARTIAL	**S9 - Doctor's personal time after a no-prescription visit**
Today	With no prescription there is no signal for the moment the patient left. Until the next click, or 15 min without CMED activity, a phone call in the room is recorded into that patient's file.
Evidence	By design; silence does not reveal a phone call
Consequence	Up to 15 min of the doctor's own time in a follow-up's file
Fix	Pause button for personal calls; trim trailing silence on the server *(effort: half a day)*

PARTIAL	**S10 - Cases the rules cannot settle**
Today	A follow-up shorter than 60 s is merged into the next patient (R3); a glance with no return signal becomes its own part; one patient in two separate parts. Each is kept, labelled by the best rule and flagged: *labelling_review* on the dashboard.
Evidence	backend tests/test_labelling.py (R5 cases)
Consequence	A person must look at the flagged file
Fix	Review list on the dashboard *(effort: half a day)*

HANDLED	**S11 - API 3b arrives after the files were archived**
Today	The rules run again. If the cut changes, the archive rejoins the parts, checks the result against the signed hash, cuts again and replaces the files; *relabelled_after_archive* is raised.
Evidence	archive_worker test_parts.py (recut, matching and mismatching hash); tests/test_labelling_server.py
Consequence	-
Fix	-

Recorder to server

GAP	**U1 - Laptop clock behind the server by more than 5 s**
Today	Grants are verified with 5 s leeway (core/crypto.py verify_grant); PyJWT rejects an *iat* in the future. Every grant fails, so every recording is held and never uploads. A clock more than 65 s ahead fails the same way (expired).
Evidence	Experiment: laptop 6 s behind -> ImmatureSignatureError; 300 s ahead -> ExpiredSignatureError
Consequence	Nothing from that PC reaches the server; audio piles up on the laptop
Fix	Verify iat/exp against the server's Date header offset (or leeway 10 min), report clock skew in the heartbeat, and have the installer enable Windows time sync *(effort: half a day)*

HANDLED	**U2 - Internet down for hours**
Today	Recording never waits; spool holds 13.5 h; authorisation retried; uploads and matching resume on reconnect.
Evidence	Live outage tests (60 s and 100 s offline) -> archived
Consequence	Delayed only
Fix	-

PARTIAL	**U3 - Server says no (unmapped clinic, revoked device, key mismatch)**
Today	Held on the laptop, asked again every 5 min, alert *authorisation_held*; nothing deleted.
Evidence	tests/test_authorisation.py (hold tests)
Consequence	Revoked device holds audio until the spool fills; then new starts are refused
Fix	Dashboard list of held sessions per PC with the reason *(effort: half a day)*

HANDLED	**U4 - What the PC saw while offline**
Today	Offline heartbeats (1/min) and every alert go to a 5 MB health log, delivered on reconnect and deleted only after the server stores them.
Evidence	Live outage test: 3 lines delivered then deleted; dashboard showed the period
Consequence	-
Fix	-

HANDLED	**U5 - Duplicate upload after a lost response**
Today	Server recognises the stored entry hash and answers duplicate.
Evidence	Segment, reassign and marker duplicate tests
Consequence	-
Fix	-

HANDLED	**U6 - A clicked patient's grant refused or invalid**
Today	The marker is still delivered (grant_jti null after the hold period) and kept in the chain; the server finds that patient's API 2 and 3b by the click's five fields instead (*marker_without_grant* / *marker_grant_invalid* alerts).
Evidence	tests/test_markers.py (grant cases), tests/test_labelling_server.py
Consequence	-
Fix	-

HANDLED	**U7 - PC installed while the clinic's internet is down**
Today	The enrolment token is tried again every minute until the server answers; a token the server refuses is not retried.
Evidence	tests/test_enrol_retry.py
Consequence	-
Fix	-

Server and database

GAP	**D1 - A migration not applied on a deployed database**
Today	Nothing checks the schema at start-up. With 013 missing every reassign got 500 (seen live); with 015 missing every click marker fails and that recording stops uploading.
Evidence	Live: chain_entries_entry_type_check violation until 013 was applied
Consequence	Uploads stall until someone notices; recorders retry for ever
Fix	A schema_version table written by each migration and checked at start (refuse to start or report degraded in /health) *(effort: half a day)*

HANDLED	**D2 - Database restarts mid-request**
Today	The request fails; the recorder and worker retry; commits are idempotent.
Evidence	Duplicate-detection tests; not re-run against a restarting database
Consequence	Delayed only
Fix	-

HANDLED	**D3 - API 2 never arrives for a recording**
Today	Recording archived 'unconfirmed' and kept; API 2 attaches whenever it arrives.
Evidence	Live: earlier TEST_NEXT and TEST_OFF archived unconfirmed
Consequence	-
Fix	-

PARTIAL	**D4 - A CMED outage makes many recordings unconfirmed**
Today	One *unconfirmed_recording* warning per recording.
Evidence	Code read
Consequence	Alert flood during every CMED outage hides real alerts
Fix	Raise one alert per clinic per outage, or only after 24 h without API 2 *(effort: 2 hours)*

HANDLED	**D5 - Background audit writer stops**
Today	Restarted on demand when dead or bound to a closed loop.
Evidence	224 backend tests pass against real PostgreSQL
Consequence	-
Fix	-

HANDLED	**D6 - The wrong patient's late records reach a relabelled file**
Today	A visit finds a recording only through the grant that recording now carries, or one of its clicks; a first-patient grant left behind by a move confirms and attaches nothing.
Evidence	tests/test_reassign.py, tests/test_labelling_server.py
Consequence	-
Fix	-

Archive, storage and backups

PARTIAL	**A1 - Server disk fills**
Today	WAVs released after a verified R2 copy and the retention period; monitor alerts at 15% free.
Evidence	archive_worker tests (122)
Consequence	If R2 is down for more than about a week the disk still fills
Fix	Release-on-pressure and an alert when copies are overdue more than 2 days *(effort: half a day)*

HANDLED	**A2 - R2 unreachable or wrong keys**
Today	/health reports the bucket state; copies are retried; *cloud_copy_overdue* after 24 h.
Evidence	tests/test_health_storage.py
Consequence	Copies delayed
Fix	-

GAP	**A3 - The droplet is lost**
Today	Audio is safe in R2. Database backups are written to a folder on the same droplet; the off-droplet copy is a manual step in the runbook.
Evidence	Code and runbook read
Consequence	Up to a week of clinical records and the catalogue lost (DigitalOcean weekly backup only)
Fix	Upload each encrypted nightly backup to its own R2 bucket *(effort: half a day)*

HANDLED	**A4 - Archive worker dies mid-copy or mid-cut**
Today	Partial files are discarded and the work restarts; a multi-part cut resumes; pieces deleted only after every part is copied and verified.
Evidence	archive_worker test_parts.py, test_worker_copy.py
Consequence	-
Fix	-

HANDLED	**A5 - A cut file does not match the signed recording**
Today	Parts are contiguous and checked to add up to every frame; rejoined, they must equal the session's signed hash before any re-cut, or the worker refuses and reports.
Evidence	Live: Sakib + Salma rejoined to the session's sha256 exactly; test_parts.py
Consequence	-
Fix	-

CMED boundary

PARTIAL	**B1 - CMED's server offline (power or internet)**
Today	API 2 and API 3 must wait in CMED's own queue; recordings are kept regardless and matched when API 2 arrives.
Evidence	Depends on CMED's queue
Consequence	Clinical data missing until CMED sends it; lost if CMED does not queue durably
Fix	Check during the CMED build session *(effort: CMED)*

HANDLED	**B2 - Same visit or prescription sent again**
Today	Identical: 200 ALREADY_RECEIVED; changed prescription: next version; previous visit fingerprinted (ALREADY_STORED / UPDATED / STORED).
Evidence	tests/test_previous_visit.py
Consequence	-
Fix	-

PARTIAL	**B3 - CMED's finished page meets the new recorder**
Today	Every reply is one their code already treats as success (RECORDING_STARTED / PROVISIONAL, SESSION_ALREADY_ACTIVE, GATE_ARMED); PATIENT_NOTED and GATE_NOT_ARMED are no longer sent. Not yet seen: whether their page sends API 2 for every click and API 1 again when the doctor goes back.
Evidence	Our reference page; tools/cmed_build_session.py written, not yet run with CMED
Consequence	If API 2 is missing for a click, that part is archived unconfirmed (kept)
Fix	Run tools/cmed_build_session.py with CMED's build *(effort: 1 session)*

HANDLED	**B4 - Unmapped or wrong clinic code**
Today	Channel B: stored with an *unmapped_clinic* alert. Recorder: CLINIC_MISMATCH, held.
Evidence	tests/test_v3_server.py
Consequence	Delayed until mapped
Fix	-

Security

PARTIAL	**X1 - Laptop stolen**
Today	Spool and the 3-second partial audio are encrypted with a DPAPI-wrapped key; the device can be revoked. Someone who can log in as that Windows user can unwrap the key.
Evidence	Code read
Consequence	Audio readable on a stolen, logged-in laptop
Fix	Require BitLocker on clinic PCs (installer check) *(effort: policy)*

HANDLED	**X2 - CMED key leaked**
Today	Revoke and issue a new key (admin endpoints).
Evidence	tests/test_v3_server.py::test_revoked_key_is_refused
Consequence	-
Fix	-

HANDLED	**X3 - Forged or replayed grant / chain entry**
Today	EdDSA only, single-use grants, signed hash chain verified per entry, markers included.
Evidence	tests/test_authorisation.py, test_chain.py
Consequence	-
Fix	-

Deployment and versions

PARTIAL	**V1 - New recorder against a server without the new endpoints**
Today	Markers are kept and retried; the recording cannot finish uploading until /session/marker (migration 015) exists.
Evidence	Code read (chain order); runbook states the order
Consequence	Uploads stall until the server is upgraded
Fix	Upgrade order: server and migrations 013-015 first, recorders after *(effort: docs)*

PARTIAL	**V2 - Old recorder builds still installed at clinics**
Today	Builds before 16 October refuse or only note a second patient, and lose up to a minute at shutdown.
Evidence	Code history
Consequence	Old PCs keep the old failure modes
Fix	Version gate: server warns per device below the current build in the dashboard; reinstall all PCs *(effort: half a day)*

HANDLED	**V3 - Recorder keys do not match the new server**
Today	Recordings are held, not deleted; tools/check_key_pair.py and runbook step 11b check it.
Evidence	Key-pair check
Consequence	Delayed until fixed
Fix	-

HANDLED	**V4 - Upgrading a PC with the installer**
Today	The installer keeps that PC's server address, CMED address and local key; the server address is set once at build time. Unused enrolment tokens last 14 days.
Evidence	Installer compiled; tools tested; upgrade not run on a real PC
Consequence	-
Fix	-

Clinic environment

PARTIAL	**E1 - Router or WiFi down (CMED's local server unreachable)**
Today	The doctor cannot open a patient, so no trigger and no recording; a session already running keeps recording.
Evidence	By design (no manual start)
Consequence	Consultations during the outage are not recorded
Fix	Ethernet to consulting-room PCs; router and access point on UPS *(effort: clinic)*

PARTIAL	**E2 - Power cut at the clinic**
Today	Laptop on battery keeps recording; server and router need UPS (see E1).
Evidence	Live outage tests
Consequence	As E1 if the router dies
Fix	UPS for router and CMED server *(effort: clinic)*

PARTIAL	**E3 - Windows update restarts the PC mid-consultation**
Today	The recording closes cleanly at shutdown (C6); after logon the consultation is resumed if within 10 min, as a second, unlinked recording (S4).
Evidence	Shutdown and resume tests
Consequence	Consultation split into two unlinked files
Fix	Set active hours / defer updates on clinic PCs; fix S4 *(effort: policy)*

HANDLED	**E4 - 28 rooms opening at once**
Today	Grant path moved off shared locks; measured earlier under the 14- and 28-room simulations.
Evidence	Earlier simulations; not re-run against the labelling changes
Consequence	-
Fix	-

PART 2

Recommended next fixes, in order

Lost or misfiled audio first, then anything that stalls silently.

Order	Item	What
1	U1	Clock skew: verify grants against the server's time, report skew, enable time sync
2	D1	Schema version check at server start (013, 014, 015)
3	B3	Run tools/cmed_build_session.py with CMED's finished build
4	A3	Nightly encrypted database backup to R2
5	S4	Tie a resumed recording to the one it continues
6	S10	Dashboard review list for flagged labels
7	S9	Trim trailing silence of no-prescription parts
8	D4	One unconfirmed alert per clinic outage, not per recording
9	C2	Segmenter survives a failed write; check free disk space
10	U3	Dashboard list of held sessions per PC

Not verified in this review

• A real USB microphone pulled out (C1), a real Jabra (C7), a real Windows shutdown (C6), a hard power cut (C4) and a real lid close (C3): each is tested through the same code with a fake device or message, not on clinic hardware.
• CMED's finished page (B3): tools/cmed_build_session.py is ready, not yet run.
• 28-room load (E4) and database restart (D2) were not re-run against today's code.
• Router, UPS and Windows-update behaviour (E1-E3) depend on each clinic's set-up.